

For my data science project, I will investigate whether the country of origin of lead actors influences movie budget, revenue, and audience ratings. Specifically, I will compare movies led by actors from Western countries with those led by actors from non-Western countries to determine if there are observable differences in financial investment and audience reception. This topic is relevant because it explores representation in the film industry using measurable outcomes.

I will use publicly available movie datasets from Kaggle, specifically the TMDb 5000 Movie Dataset, which contains metadata for approximately 5,000 movies. The dataset includes variables such as movie title, budget, revenue, vote average (rating), release date, and cast information. This provides a sufficient sample size for meaningful analysis.

To collect the necessary data, I will download the dataset and extract the first credited cast member for each movie as a proxy for the lead actor. Since actor nationality is not directly included in the dataset, I will supplement the data by identifying the country of origin of each lead actor using publicly available sources such as Wikipedia or TMDb profiles. Based on this information, I will classify actors into two groups: Western and non-Western.

After preparing the dataset, I will analyze differences between these groups using descriptive statistics and visualizations. I will compare average budget, revenue, and ratings, and present the findings using charts such as bar plots and box plots. The dataset's size and structure make it well-suited for this analysis, allowing for a clear and manageable investigation of the research question.